

Overcoming Facial Animation Hurdles by Leveraging Interdisciplinary Knowledge and AI-Assisted Creative Process

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Abstract

This paper addresses the challenge of limited acting skills in facial animation education, which is further heightened by the advent of realistic digital humans such as Unreal Engine MetaHuman. We introduce a transformative interdisciplinary framework that combines animation technology (Unreal Engine MetaHuman) with psychological theory (Scherer's appraisal theory) to address this issue. Together with the integration of AI tools in the workflow, such as Microsoft Copilot, Adobe Firefly, and an AI voice changer, the framework supports student creativity, helping students overcome performance anxiety and develop nuanced facial animation performance.

CCS Concepts

• **Applied computing** → Arts and humanities; Media arts; • **Computing methodologies** → Computer graphics.

ACM Reference Format:

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1 Introduction

Facial animation requires technical skill and emotional depth. However, hyper-realistic avatars like MetaHuman (Epic Games, 2024) heighten uncanny valley risk (Mori, 1970). Without acting training (Hooks, 2023), students struggle with performance anxiety and authentic expression in demanding animation courses. This project presents a transformative framework grounded in interdisciplinary practice, using Scherer's appraisal theory of emotion (Scherer, 1984) to help students systematically analyze emotional states and move beyond superficial mimicry. Using generative AI tools for monologue design, episodic visuals, and dialogue voices, the framework aids empathetic understanding, authentic performance, and lower anxiety, serving as a creative partner for students with limited acting experience.

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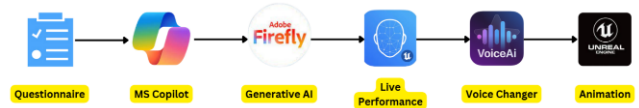


Figure 1: Overview of the Interdisciplinary Framework with AI-Assisted Creative Process

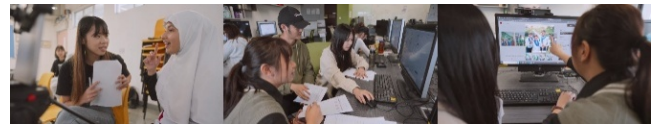


Figure 2: A student interviews an ethnic minority peer using the Geneva Appraisal Questionnaire to collect emotional experiences for monologue creation (left). MS Copilot is used to analyze the responses to shape the monologue's structure and tone (center). Adobe Firefly generates episodic memory visuals from the monologue to enhance the performance (right).

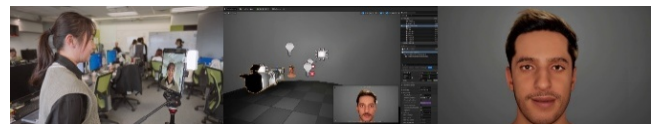


Figure 3: The student's live monologue is recorded using Live Link Face on an iPad Pro (left). The performance data is then retargeted to a MetaHuman with proper staging and lighting (center). The final facial animation features an AI voice-converted performance for a dialogue of a different gender character (right).

2 The Framework

The framework consists of different stages in which appraisal theory and AI tools are used, respectively. Through a structured and multi-stage workflow, students integrate psychological modelling of emotional experiences with AI tools, from monologue design to facial animation performance. Figure 1 illustrates how appraisal theory and artificial intelligence are incorporated throughout each stage to create genuine facial animation using MetaHuman (see Figure 2 for design steps; see Figure 3 for performance steps).

Table 1: Pre-survey questions and results summary

Questions summary	Results summary
The pre-survey questions focused on students' understanding of empathy, their ability to interpret emotions from facial expressions, their cross-cultural communication skills, and their expectations for learning and impact through the service-learning project.	Sixteen students participated in the pre-survey. The results show that students understood empathy as emotional support, active listening, and perspective-taking. They demonstrated awareness of interpreting emotions from facial expressions and regularly sought to understand people from different cultural backgrounds through conversation, media, and events.

2.1 Interdisciplinary Knowledge: Applying Scherer's Appraisal Theory of Emotion

Students first start by applying the Geneva Appraisal Questionnaire (GAQ) (Geneva Emotion Research Group, 2015) to interview ethnic minority peers and record emotional episodes, systematically investigating factors like novelty, pleasantness, goal conduciveness, and coping potential (Figure 2). These structured reflections generate genuine narrative material and enhance students' empathetic understanding, anchoring facial animation in real emotional experiences.

2.2 AI-Assisted Creative Process: Deepening Engagement Through Embodiment

Microsoft Copilot assists students in transforming appraisal-based reflections into emotionally rich monologues by breaking down emotional beats for deeper internalization. Adobe Firefly visualizes these narratives into vivid episodic scenes, enhancing character portrayal in facial animation (Figure 2). The process then progresses to embodiment through live capture, retargeting, and voice adaptation, creating a seamless digital performance (Figure 3).

3 Results

The FILM4075 Facial Animation Service-Learning Project at Hong Kong Baptist University implemented the proposed framework across two cohorts (2023–2024), pairing 28 undergraduates with 33 ethnic minority (EM) secondary students to create 56 MetaHuman facial animations about ethnic minority students' emotional experiences (HKBU, 2024). Pre- and post-course surveys (Lee, 2024) were administered to the second cohort of 17 students in September 2024 and December 2024, respectively, to examine baseline attitudes and subsequent shifts following the service-learning facial animation project. Questions and results of both surveys are summarized in Table 1 and Table 2, respectively.

Table 2: Post-survey questions and results summary

Questions summary	Results summary
The post-survey questions asked students how their definition and experience of empathy changed after the project, how their compassion for ethnic minority communities evolved, how the service-learning affected their understanding of cross-cultural exchange, and whether their empathy towards ethnic minorities in Hong Kong increased after participating in the course.	Ten students participated in and completed the post-survey. The results indicate that most participants reported an increased understanding and practice of empathy, especially towards ethnic minority communities. Students gained deeper compassion, challenged stereotypes, and recognized the importance of cross-cultural exchange for both personal growth and future professional endeavors.

4 Discussion

The interdisciplinary, appraisal-driven, and AI-assisted workflow successfully lowered the barrier for novice animators, enabling credible facial animation performances without prior acting experience. Collaboration with ethnic minority participants helped anchor these animations in authentic emotional narratives. However, limited direct input and involvement from ethnic minority students during production restricted the representation of their perspectives and the opportunity for feedback, pointing to a significant area for improvement. The use of generative AI contributed positively to creative development, but also introduced risks of bias and generic outcomes, which may have affected the authenticity of the results. Addressing these issues requires more inclusive production pipelines, with expanded opportunities for ethnically diverse students to contribute throughout all stages and robust feedback mechanisms to ensure authentic representation. These points will guide future iterations of the project as greater cultural participation and enhanced review processes will be essential for advancing educational and creative outcomes.

References

- Epic Games. 2024. MetaHuman | Digital Humans—Epic Games. <https://www.metahuman.com/en-US>
- Geneva Emotion Research Group. 2015. Research Material—Swiss Center For Affective Sciences—UNIGE. <https://www.unige.ch/cisa/research/materials-and-online-research/research-material/>
- HKBU. 2024. HKBU Service-learning Facial Animation Project—YouTube. <https://www.youtube.com/@SLFacialAnimProj>
- Ed Hooks. 2023. Acting for Animators (5th ed.). Routledge. <https://doi.org/10.4324/9781003289753>
- Kelvin Lee. 2024. Pre- and Post surveys for Service-learning Project.
- Masahiro Mori. 1970. Uncanny Valley. *Energy*, 7, 33–35.
- Klaus Scherer. 1984. On the Nature and Function of Emotion: A Component Process Approach. In *Approaches To Emotion*. Psychology Press.